CricketLens - Stage 2, Session 02

Deeper Exploratory Data Analysis | Real men's IPL data | 18 October 2026

Development status: Stage 2 in progress. This report is an evidence-based interim EDA chapter, not a final ML evaluation. All calculations exclude 2026, held for later model testing.

Questions investigated

How does innings length change typical scores? Do scoring phases differ within the same innings? Are venue averages robust to changing season mix? Is a toss decision associated with results? Where do missing city values occur?

Verified numerical findings

Observation	2018-2025 historical result
All recorded first innings	n=533; mean 175.80 runs
First innings with 120 legal balls	n=501; mean 178.88 runs
First innings shorter than 120 legal balls	n=32; mean 127.50 runs
Powerplay vs death, same innings	n=937 paired innings; difference +2.26 runs per 6 balls
Paired difference, 95% bootstrap interval	+2.02 to +2.51, resampling matches

Methodological guardrails

These comparisons describe historical records; they do not isolate causal pitch, venue or toss effects. For late-over comparisons we select innings that reached the death phase; this excludes early-ending innings. Venue comparisons include full-length first innings only, and season-centering does not control teams, weather or matchup composition.

1. Innings length and within-innings scoring

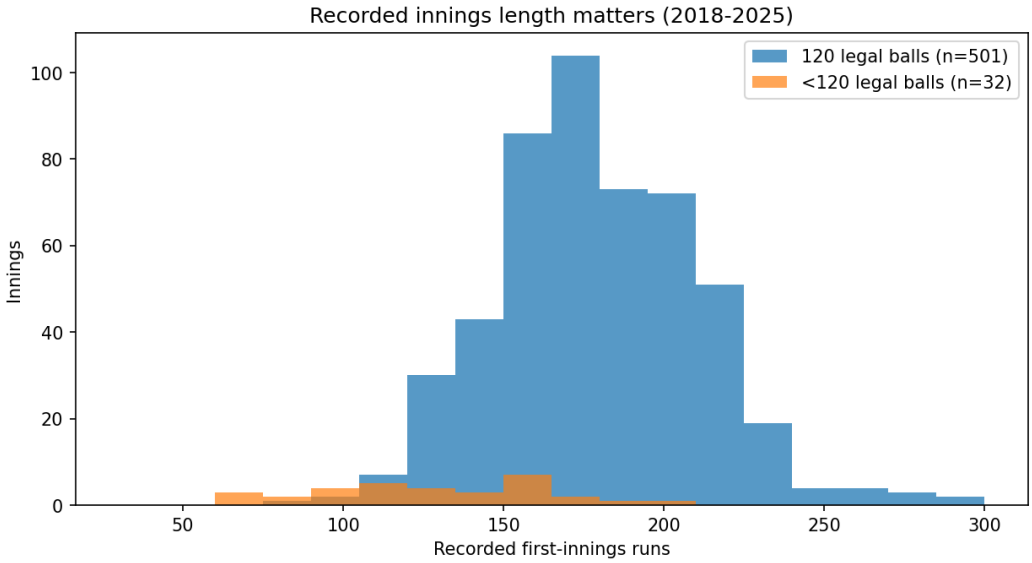


Figure 12. Full 120-ball vs shorter first-innings score distributions. Both categories retain genuine match outcomes.

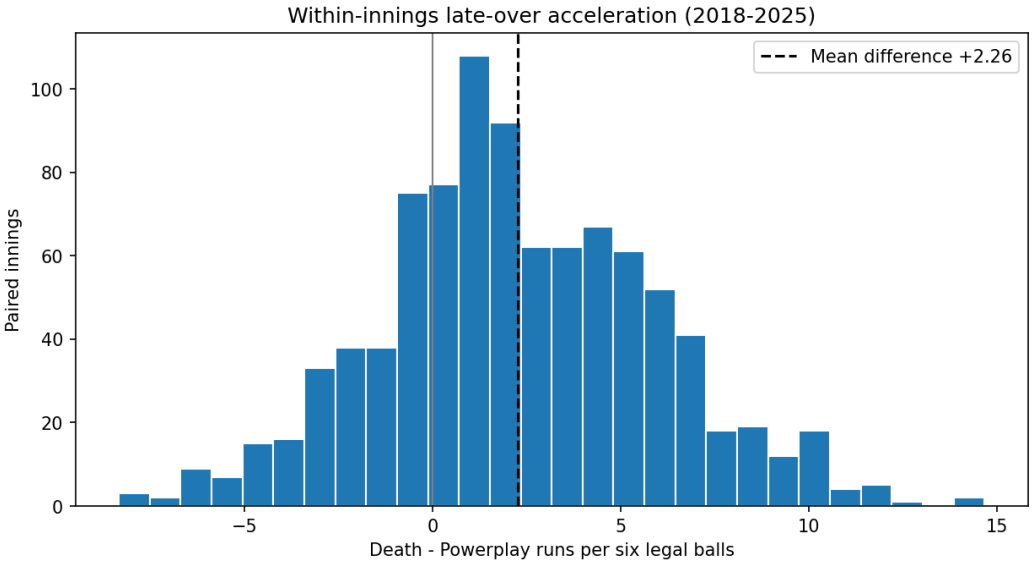


Figure 14. Death minus Powerplay scoring rates computed within the same qualifying innings.

The overall mean of 175.80 is an average of unlike innings durations. Restricting the denominator to 120-ball innings raises it to 178.88; the shorter 32 innings average 127.50. Conditional selection is explicit, not a claim that complete innings cause higher batting quality.

2. Venue and era-aware comparisons

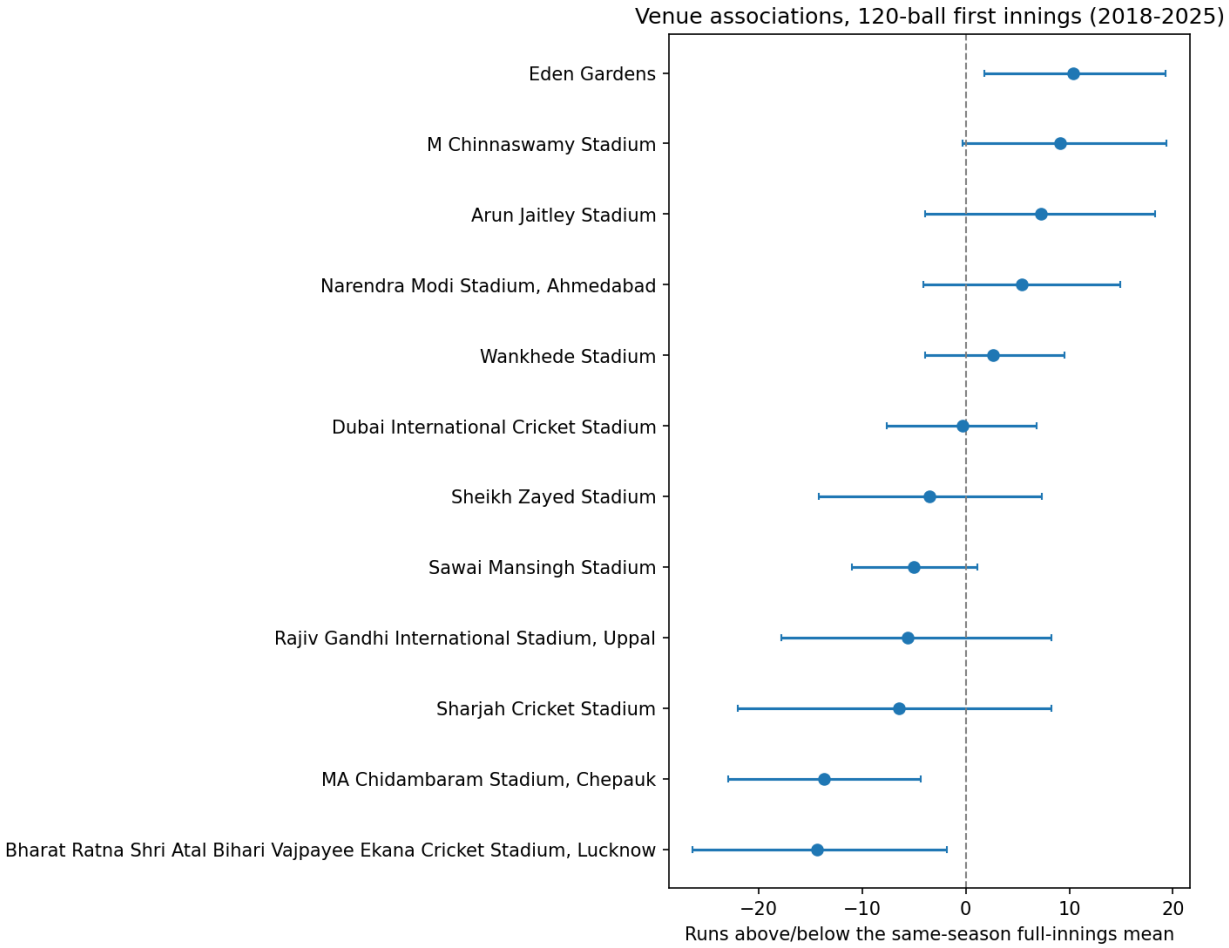


Figure 13. Venue mean score minus its season-wide mean for full-length first innings; 95% within-venue bootstrap intervals.

Measured examples: Eden Gardens: +10.34 runs (n=37); MA Chidambaram Stadium, Chepauk: -13.71 (n=38); Ekana Stadium, Lucknow: -14.32 (n=21). These are season-centered associations, not causal venue advantages. Minimum 20 qualifying innings per venue; 12 venues meet that threshold.

Bootstrap intervals quantify observation-to-observation spread under a resampling model. They do not adjust for unmeasured factors or prevent selection bias.

3. Categorical EDA and missingness

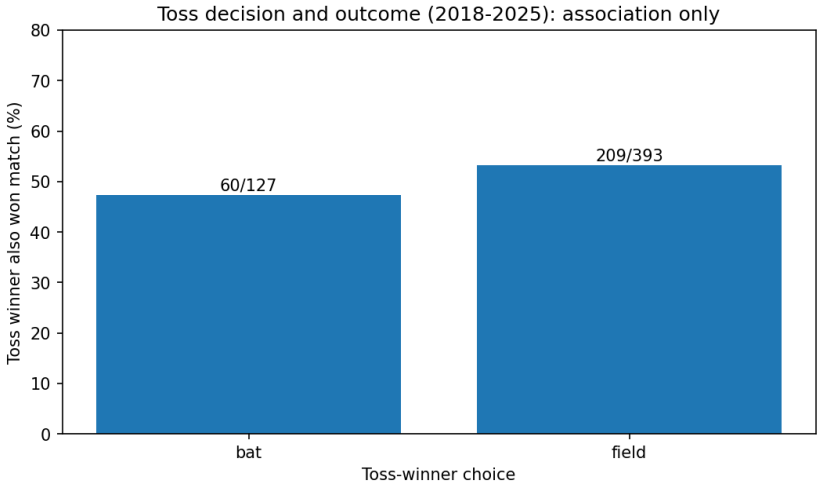


Figure 16. Among decided matches, toss winner match-win percentages grouped by toss decision.

Toss winners choosing to bat won 60/127 (47.24%) recorded decided matches; choosing to field won 209/393 (53.18%). Fisher exact two-sided p=0.262. We should not claim a toss-choice advantage from this observational crosstab.

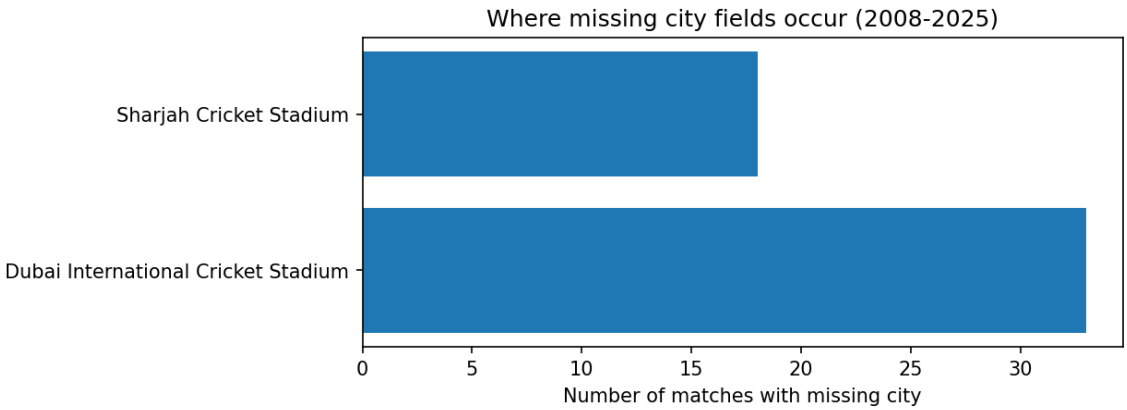


Figure 17. Missing city values by observed venue in IPL match records before 2026.

Of 1,169 matches before 2026, 51 lack a city. Those 51 are confined to Dubai (33) and Sharjah (18) in the observed data. This location dependence suggests considering observed missingness patterns, but cannot establish MCAR, MAR or MNAR definitively.

4. Syllabus alignment, limitations, next actions

Skills demonstrated

Descriptive measures (mean, median, trimmed mean, spread and Tukey fences); univariate histograms; paired bivariate contrasts; a contingency table; Fisher exact test; stratified/season-centered comparisons; bootstrap uncertainty; missingness visualization; explicit variable measurement scales and sampling-bias discussion.

Important caveats

• The source is a historical IPL record, not a randomized cricket population sample; restriction to 2018-2025 and full 20-over innings changes what the results describe.

• Paired phase contrasts require innings to last into death overs. Bootstrap intervals for phase differences resample whole matches, but do not correct for all sources of bias.

• Toss outcomes and venue differences are observational; no causal claims or adjusted risk estimates are implied.

• Missingness mechanisms cannot be established from a summary table. Missing outcomes and structural not-applicable fields require domain-aware handling.

• No 2026 data has been used. These are not deployable ML features or final model test metrics.

Exact reproducibility

Source tables: data/processed/ from Stage 1. Python entry point: `python -m cricketlens.stage02_deep_eda`. Tables: reports/stage_02/session_02/; charts: reports/figures/stage_02/ (Figures 12-17). Code tests: tests/test_stage02_deep_eda.py. Detailed explanations: docs/STAGE_02_SESSION_02_FINDINGS.md.

Next

Combine EDA Sessions 01 and 02 into a coherent course-aligned chapter, review interpretations with the professor, select figures for the six-tab dashboard, then sign off Stage 2. Stage 3 model development remains separate.

Source: Cricsheet men's IPL JSON archive, processed with CricketLens scripts. Disclose AI assistance as required by the course instructions.